www.tntextbooks.in

The electric potential energy difference can be written as $\Delta U = q' \Delta V$. Physically potential difference between two points is a meaningful quantity. The value of the potential itself at one point is not meaningful. Therefore the point R is taken at infinity and its potential is considered as zero $(V_\infty = 0)$.

Then **the electric potential at a point P is equal to the work done by an external force to bring a unit positive charge with constant velocity from infinity to the point P in the region of the external electric field $\vec{E}$.** Mathematically this is written as

$$V_P = -\int_\infty^P \vec{E} \cdot d\vec{r} \qquad (1.31)$$

Important points

1. Electric potential at point P depends only on the electric field which is due to the source charge q and not on the test charge q′. Unit positive charge is brought from infinity to the point P with constant velocity because external agency should not impart any kinetic energy to the test charge.
2. From equation (1.29), the unit of electric potential is Joule per coulomb. The practical unit is volt (V) named after Alessandro Volta (1745-1827) who invented the electrical battery. The potential difference between two points is expressed in terms of voltage.

1.5.2 Electric potential due to a point charge

Consider a positive charge q kept fixed at the origin. Let P be a point at distance r from the charge q. This is shown in Figure 1.23.

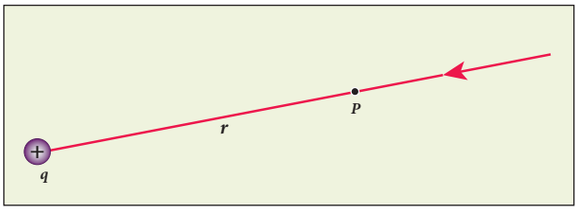


Figure 1.23 Electrostatic potential at a point P

The electric potential at the point P is

$$V = \int_\infty^r \left(-\vec{E}\right) \cdot d\vec{r} = -\int_\infty^r \vec{E} \cdot d\vec{r} \qquad (1.32)$$

Electric field due to positive point charge q is

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$$

$$V = \frac{-1}{4\pi\varepsilon_0}\int_\infty^r \frac{q}{r^2}\hat{r}.d\vec{r}$$

The infinitesimal displacement vector, $d\vec{r} = dr\hat{r}$ and using $\hat{r}.\hat{r} = 1$, we have

$$V = -\frac{1}{4\pi\varepsilon_\circ}\int_\infty^r \frac{q}{r^2}\hat{r}\cdot dr\hat{r} = -\frac{1}{4\pi\varepsilon_\circ}\int_\infty^r \frac{q}{r^2}dr$$

After the integration,

$$V = -\frac{1}{4\pi\varepsilon_\circ}q\left\{-\frac{1}{r}\right\}_\infty^r = \frac{1}{4\pi\varepsilon_\circ}\frac{q}{r}$$

Hence the electric potential due to a point charge q at a distance r is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad (1.33)$$

Important points

(i) If the source charge q is positive, V > 0. If q is negative, then V is negative and equal to $V = -\frac{1}{4\pi\varepsilon_0}\frac{q}{r}$

(ii) The description of motion of objects using the concept of potential or potential energy is simpler than that using the concept of field.

UNIT-1(XII-Physics_Vol-1).indd 28 04-03-2019 10:25:31